

L2: Computational Formalization of Semiotic Density

Operationalizing $\sigma(t)$ in Embedding Space

(Appendix for: *Semiotic Density: From Ideographic Writing to Framework Injection*)

Renato Aparecido Gomes

Working Draft — April 2026

Motivation

The main paper defines semiotic density as:

$$\sigma(t) = \frac{|\mathcal{F}(t)|}{|t|_{\text{surface}}}$$

where $\mathcal{F}(t)$ is the set of frames, prototypes, and inferential pathways activated by t . This definition is theoretically grounded but computationally underspecified: $|\mathcal{F}(t)|$ has no operational procedure for measurement.

Without operationalization, two problems follow:

1. **L6 is blocked:** the claim that FI improves output quality through density cannot be verified quantitatively without a density measure.
2. **Design guidance remains intuitive:** practitioners cannot compare terms or predict which framework construction achieves higher density.

This appendix provides: (a) an embedding-space proxy for $\sigma(t)$, (b) an estimable quantity $\hat{\sigma}_{\text{emb}}(t)$ computable from any sentence encoder, (c) formal characterizations of the five SDE operations as transformations in the embedding manifold, and (d) testable predictions that connect the formalization to existing experimental data (D1, A1).

1 The Embedding Space Proxy

1.1 Setup

Let $\mathbf{E} : \mathcal{V}^* \rightarrow \mathbb{R}^d$ be a sentence encoder (e.g., `text-embedding-3-large` or any dense retrieval model) that maps a string to a d -dimensional embedding.

For a term $t \in \mathcal{V}^*$, define its **implicit semantic neighborhood** as the set of concepts co-activated with t in context:

$$\mathcal{N}_\epsilon(t) = \{c \in \mathcal{C} \mid \cos(\mathbf{E}(t), \mathbf{E}(c)) \geq \epsilon\}$$

where $\mathcal{C}$ is a reference concept vocabulary and ϵ is a threshold (empirically, $\epsilon \approx 0.6$ works well for domain terms).

Definition 1.1 (Embedding-Space Semiotic Density). *The **embedding-space semiotic density** of a term t at threshold ϵ is:*

$$\hat{\sigma}_\epsilon(t) = \frac{|\mathcal{N}_\epsilon(t)|}{|t|_{\text{tokens}}}$$

where $|t|_{\text{tokens}}$ is the number of tokens in t under the model’s tokenizer.

This definition operationalizes $|\mathcal{F}(t)|$ as the number of concepts within cosine distance ϵ in embedding space. The approximation is justified by the distributional hypothesis: a term’s embedding encodes its co-occurrence statistics, which are the statistical signature of frame activation.

1.2 Properties

Proposition 1.1 (Density ordering reflects intuition). *Under Definition ??, for typical domain terms:*

$$\hat{\sigma}_\epsilon(\text{"selfie"}) > \hat{\sigma}_\epsilon(\text{"photograph"}) > \hat{\sigma}_\epsilon(\text{"image of a person"})$$

Remark 1.1. *This ordering is empirically verifiable: “selfie” should yield a denser neighborhood than “photograph” because its distributional context (social media, front-facing camera, arm extension, youth culture) is more structured and coherent than the diffuse context of “photograph.” The compactness of the neighborhood — its internal coherence — is also relevant; see Section ??.*

Proposition 1.2 (Density-Quality Hypothesis). *For a framework $\mathbf{F} = (t_1, \dots, t_k)$, let:*

$$\bar{\sigma}(\mathbf{F}) = \frac{1}{k} \sum_{i=1}^k \hat{\sigma}_\epsilon(t_i)$$

Then output quality $Q(\mathbf{F})$ is positively correlated with $\bar{\sigma}(\mathbf{F})$ when task complexity exceeds a threshold:

$$\text{Corr}(Q(\mathbf{F}), \bar{\sigma}(\mathbf{F})) > 0 \quad \text{for complex analytical tasks}$$

This is the central testable claim of L6. It connects the theoretical density measure to the empirical output quality data from D1 and A1.

2 Neighborhood Entropy as Coherence Measure

Not all large neighborhoods are equally valuable. A term that activates many incoherent, unrelated concepts is not semiotically dense in the productive sense — it is merely ambiguous.

Definition 2.1 (Neighborhood Entropy). *The **neighborhood entropy** of t at threshold ϵ is:*

$$H_\epsilon(t) = - \sum_{c \in \mathcal{N}_\epsilon(t)} p(c | t) \log p(c | t)$$

where $p(c | t) \propto \cos(\mathbf{E}(t), \mathbf{E}(c)) - \epsilon$ (similarity minus threshold, normalized).

Low entropy = coherent, focused activation (dense in the productive sense).
High entropy = diffuse, unstructured activation (noisy density).

Definition 2.2 (Structured Semiotic Density). *The **structured semiotic density** of t is:*

$$\sigma^*(t) = \frac{|\mathcal{N}_\epsilon(t)|}{|t|_{\text{tokens}}} \cdot \frac{1}{H_\epsilon(t) + 1}$$

The $(H + 1)^{-1}$ term penalizes incoherent neighborhoods: a term with many semantically distant activations receives lower structured density than one with fewer but tightly clustered activations. This distinguishes *selfie* (coherent frame activation) from *thing* (large but incoherent).

Proposition 2.1 (UID Condition). *A framework $\mathbf{F} = (t_1, \dots, t_k)$ satisfies the Uniformity of Density principle iff:*

$$\text{Var}(\sigma^*(t_1), \dots, \sigma^*(t_k)) < \delta_{UID}$$

for an empirically calibrated threshold δ_{UID} .

This provides an actionable check: compute σ^* for all key terms in a framework and verify that the variance is below threshold.

3 SDE Operations as Embedding Transformations

The five SDE operations defined linguistically in the main paper correspond to well-defined transformations in embedding space. This correspondence is what makes SDE formally tractable.

Let $\mathbf{v}(t) = \mathbf{E}(t) \in \mathbb{R}^d$ denote the embedding of t .

3.1 Densify

Linguistic: Replace t_{thin} with t_{dense} that activates a richer semantic field.

Embedding: Find $t' = \arg \max_{t \in \mathcal{V}^*} \sigma^*(t)$ subject to $\cos(\mathbf{v}(t'), \mathbf{v}(t_{\text{thin}})) \geq \alpha$ (preserve intended meaning direction) and $|t'|_{\text{tokens}} \leq |t_{\text{thin}}|_{\text{tokens}}$.

Geometric interpretation: Move to a nearby point in embedding space with a larger, more coherent neighborhood.

3.2 Rarefy

Linguistic: Remove unwanted associations from $t_{\text{misleading}}$.

Embedding: Let $\mathbf{u}$ be the mean embedding of the unwanted concepts $\mathcal{U}$. Compute:

$$\mathbf{v}' = \mathbf{v}(t) - \lambda \cdot \text{proj}_{\mathbf{u}} \mathbf{v}(t)$$

where $\text{proj}_{\mathbf{u}}$ is projection onto $\mathbf{u}$. This is STEER's subtraction operation [?] applied to noise dimensions.

Geometric: Step away from the unwanted subspace while preserving the useful dimensions.

3.3 Rotate

Linguistic: Preserve the structural role of t but shift its semantic domain from D_1 to D_2 .

Embedding: Let μ_{D_1} and μ_{D_2} be the centroid embeddings of domains D_1 and D_2 respectively:

$$\mathbf{v}' = \mathbf{v}(t) - \mu_{D_1} + \mu_{D_2}$$

This is the analogy operation $\mathbf{v}(\text{king}) - \mathbf{v}(\text{man}) + \mathbf{v}(\text{woman}) \approx \mathbf{v}(\text{queen})$ [?] generalized to domain centroids.

Geometric: Parallel transport in embedding space from domain D_1 to domain D_2 .

3.4 Subtract

Linguistic: Remove dimension d from $\mathcal{F}(t)$.

Embedding: Let $\mathbf{d}$ be the direction in embedding space corresponding to dimension d (obtainable via contrastive pairs). Then:

$$\mathbf{v}' = \mathbf{v}(t) - (\mathbf{v}(t) \cdot \hat{\mathbf{d}}) \hat{\mathbf{d}}$$

This removes the component of $\mathbf{v}(t)$ along direction $\hat{\mathbf{d}}$.

3.5 Blend

Linguistic: Compose t_1 and t_2 to produce emergent meaning.

Embedding: A simple linear blend $(\mathbf{v}(t_1) + \mathbf{v}(t_2))/2$ does not capture emergence. Instead, emergence corresponds to the component of the composition *orthogonal* to both input embeddings:

$$\mathbf{v}_{\text{emergent}} = \mathbf{v}(t_1 \oplus t_2) - \text{proj}_{\mathbf{v}(t_1)} \mathbf{v}(t_1 \oplus t_2) - \text{proj}_{\mathbf{v}(t_2)} \mathbf{v}(t_1 \oplus t_2)$$

A productive blend has $\|\mathbf{v}_{\text{emergent}}\| > 0$ — the composed phrase occupies a region of embedding space not spanned by either component.

Definition 3.1 (Blend Productivity). *The **blend productivity** of (t_1, t_2) is:*

$$\beta(t_1, t_2) = \|\mathbf{v}_{emergent}(t_1 \oplus t_2)\|_2$$

Higher β = more emergent content = more productive blend.

Empirical prediction: “selfie $\oplus$ giraffe” should have higher β than “selfie $\oplus$ person”, because the former crosses domain boundaries (human social practice $\times$ animal biology) while the latter remains within the same frame.

4 The SD-Score Algorithm

The SD-Score algorithm ranks candidate terms for framework construction by structured semiotic density.

Input:

- Target semantic field $\mathcal{T}$ (described as a set of seed concepts)
- Candidate term set $\mathcal{C}$
- Embedding model $\mathbf{E}$, threshold ϵ , penalty weight λ

Output: Ranked list of $(t, \sigma^*(t), H_\epsilon(t))$ triples.

Algorithm SD-Score:

1. Compute $\boldsymbol{\mu}_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \mathbf{E}(s)$ // field centroid
2. For each $t \in \mathcal{C}$:
 - (a) Compute $\mathbf{v}(t) = \mathbf{E}(t)$
 - (b) Compute field alignment: $a(t) = \cos(\mathbf{v}(t), \boldsymbol{\mu}_{\mathcal{T}})$
 - (c) Compute neighborhood: $\mathcal{N}_\epsilon(t)$ from reference vocabulary
 - (d) Compute entropy: $H_\epsilon(t)$ per Definition ??
 - (e) Compute $\sigma^*(t)$ per Definition ??
 - (f) Compute final score: $\text{SD}(t) = a(t) \cdot \sigma^*(t)$
3. Return top- k terms ranked by $\text{SD}(t)$

The $a(t)$ term ensures that only terms aligned with the target semantic field receive high scores — preventing high-density off-topic terms from dominating.

5 Testable Predictions Against Experimental Data

The formalization generates predictions testable against the existing D1 and A1 experimental datasets.

5.1 Prediction 1 — FI Terms Rank Higher in SD-Score

Claim: The key terms in the D1 framework injections (e.g., “regulatory compliance,” “anticipatory breach,” “epistemic calibration”) should score higher on SD-Score than the corresponding `pe_baseline` terms (e.g., “be accurate,” “be professional,” “respond precisely”).

Test: Extract key terms from all FI and `pe_baseline` system prompts in A1. Compute SD-Score for each term against the domain’s target semantic field. Test whether $\overline{SD}(\text{FI terms}) > \overline{SD}(\text{PE terms})$.

Expected outcome: FI terms score significantly higher, providing a mechanistic account of why FI outperforms PE.

5.2 Prediction 2 — `fi_avaliativo` Has Highest Density

Claim: Among the three FI types (declarative, procedural, evaluative), `fi_avaliativo` should have the highest mean σ^* of its key terms.

Rationale: Evaluative FI uses quality criterion terms (“precisão técnica,” “completude relevante,” “calibração”) that activate structured domain frames more coherently than declarative identity terms or procedural step labels.

Test: Compute SD-Score distribution for each FI type’s key terms across A1 domains. Test $\bar{\sigma}^*(\text{fi_avaliativo}) > \bar{\sigma}^*(\text{fi_declarativo}), \bar{\sigma}^*(\text{fi_procedural})$.

Expected outcome: Consistent with A1 finding that `fi_avaliativo` is the most robust type across task types and models.

5.3 Prediction 3 — Thinking Token Count Correlates with $\bar{\sigma}^*$

Claim: From D1, the FI frameworks that generated more thinking tokens (`fi_extended` vs. `no_fi_extended`: 566 vs. 404) correspond to higher σ^* values in the framework’s key terms.

Test: For each D1 scenario, compute $\bar{\sigma}^*$ of the FI’s key terms. Correlate with the observed thinking token count.

Expected outcome: $\text{Corr}(\bar{\sigma}^*, \text{thinking_tokens}) > 0$, providing a mechanistic bridge between semiotic density and the epistemic prior function identified in the D1 paper.

5.4 Prediction 4 — GPT-4o Gap Partially Explained by SD Mismatch

Claim: The degradation of `fi_procedural` in GPT-4o (A1: $d = -1.49$) corresponds to lower σ^* of procedural terms in GPT-4o’s embedding space relative to Claude’s.

Test: Compute SD-Score for `fi_procedural` key terms using both a Claude-aligned encoder and a GPT-4o-aligned encoder (or the model’s own embeddings if accessible). Test if the gap in density scores predicts the gap in output quality.

Expected outcome: Procedural terms (“PASSO 1,” “DECOMPOSIÇÃO,” “execute nesta ordem”) have lower coherent neighborhood density in GPT-4o’s representation space, explaining why `fi_procedural` is the most fragile type cross-model.

6 Implementation Sketch

A minimal implementation of SD-Score requires:

1. **Embedding model:** Any high-quality sentence encoder. Recommended: `text-embedding-3-large` (OpenAI) or `voyage-3` (Anthropic) for consistency with the generation models used in D1/A1.
2. **Reference vocabulary $\mathcal{C}$:** A domain concept lexicon. For L6 validation, use the task prompts themselves as the reference set (seed concepts extracted by noun phrase chunking).

3. **Threshold ϵ :** Calibrate empirically. A grid search over $\epsilon \in \{0.5, 0.6, 0.7, 0.8\}$ with L6 outcome as criterion will identify the optimal threshold.
4. **Tokenizer:** Use the same tokenizer as the generation model for $|t|_{\text{tokens}}$.

The full implementation (150 lines of Python) is planned as `compute_sd_score.py` in the `fi-sandbox-benchmark` repository. It will require no dependencies beyond `anthropic`, `numpy`, and `scikit-learn`.

7 Connection to STEER

STEER [?] provides 16 algebraic operations on embedding vectors — including NAVIGATE, ROTATE, SUBTRACT, and BLEND — which map directly to the SDE operations formalized above.

The relationship is complementary:

SDE Operation	STEER Operation	Level
Densify	Navigate(target_field)	Retrieval
Rarefy	Subtract(unwanted_direction)	Embedding manipulation
Rotate	Rotate(source_domain, target)	Domain shift
Subtract	Subtract(dimension)	Embedding manipulation
Blend	Blend(t1, t2) + emergence test	Composition

SDE provides the *linguistic decision* (which operation to apply, which terms to target); STEER provides the *geometric implementation* (how to compute the transformation in embedding space). Together they form a complete pipeline: linguistic analysis $\rightarrow$ embedding computation $\rightarrow$ optimized framework construction.

8 Open Problems

1. **Context-dependence of $\mathcal{N}_\epsilon(t)$:** The neighborhood of a term shifts with context. “Compliance” in a legal context activates different neighbors than in a software context. A context-aware density measure would compute $\sigma^*(t \mid \text{context})$ using a contextual encoder rather than a context-free sentence embedding.
2. **Threshold calibration:** The choice of ϵ significantly affects σ^* values. An optimal ϵ may be domain-specific. A calibration procedure using downstream task performance (Prediction 1–4) would ground ϵ empirically.
3. **Interaction density:** The density of a multi-term framework is not simply the mean of individual σ^* values. Two highly dense terms may interact destructively (competing frames) or constructively (productive blend). A pairwise interaction term:

$$I(t_i, t_j) = \beta(t_i, t_j) - \frac{\sigma^*(t_i) + \sigma^*(t_j)}{2}$$

captures whether the blend adds or subtracts from individual densities.

4. **Density decay with context length:** As the context window fills, early framework terms lose influence. A temporal density function $\sigma^*(t, \ell)$ that models decay as a function of position ℓ from the current generation point would enable optimal framework placement strategies.
5. **Cross-model density invariance:** Prediction 4 suggests that σ^* values may differ across embedding spaces. Understanding when and why density values transfer across models is essential for cross-model FI design — directly relevant to A1’s GPT-4o findings.

Integration Notes (for DS paper revision)

Where to integrate:

- Definitions ?? and ?? : Insert as new subsection in Section 2 of DS paper, after the current Definition 1.
- Propositions ?? and ?? : Add as Propositions 2.1 and 2.2 in the DS paper.
- Sections ?? (embedding transformations): Merge with Section 6 of DS paper as a subsection “*Formal Characterization*” after each operation’s linguistic definition.
- Section ?? : Add as Appendix B in DS paper, titled “*Computational Predictions*”.
- Section ?? : Add as pseudo-code box in Section 6 or as Appendix B.

What this unlocks for L6: With SD-Score implemented, L6 becomes a regression study: compute $\bar{\sigma}^*(\mathbf{F})$ for all frameworks in A1 and D1, and regress against observed output quality. If $\beta > 0$ (Proposition ??), the density-quality hypothesis is empirically confirmed and the DS paper gains an experimental section.

References

- [1] Gomes, R. A. (2026). STEER: Sixteen algebraic operations for embedding-space retrieval. *Zenodo*. DOI: 10.5281/zenodo.19325724.
- [2] Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S., & Dean, J. (2013). Distributed representations of words and phrases and their compositionality. *Advances in Neural Information Processing Systems*, 26, 3111–3119.
- [3] Gomes, R. A. (2025). Framework injection: A structured approach to instantiating expert cognitive identities in large language models. *Working paper*. [Pending Zenodo DOI].